RETAILPULSE: *AI-POWERED CUSTOMER ANALYTICS & DEMAND FORECASTING*

Transforming Retail Intelligence with AI-Driven Insights.

ZIDIO INTERNSHIP PROJECT

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RETAILPULSE TURNS SALES DATA INTO DECISIONS

RetailPulse turns sales, customer, product and inventory data into business insights using machine learning and interactive visualization, built on synthetic retail datasets.

It combines customer segmentation, demand forecasting and inventory analytics in an interactive dashboard built with Python and Streamlit.

Target Users / Use Cases

- Retail store managers
- E-commerce businesses
- Inventory and supply chain teams
- Business analysts

Vision & Objectives

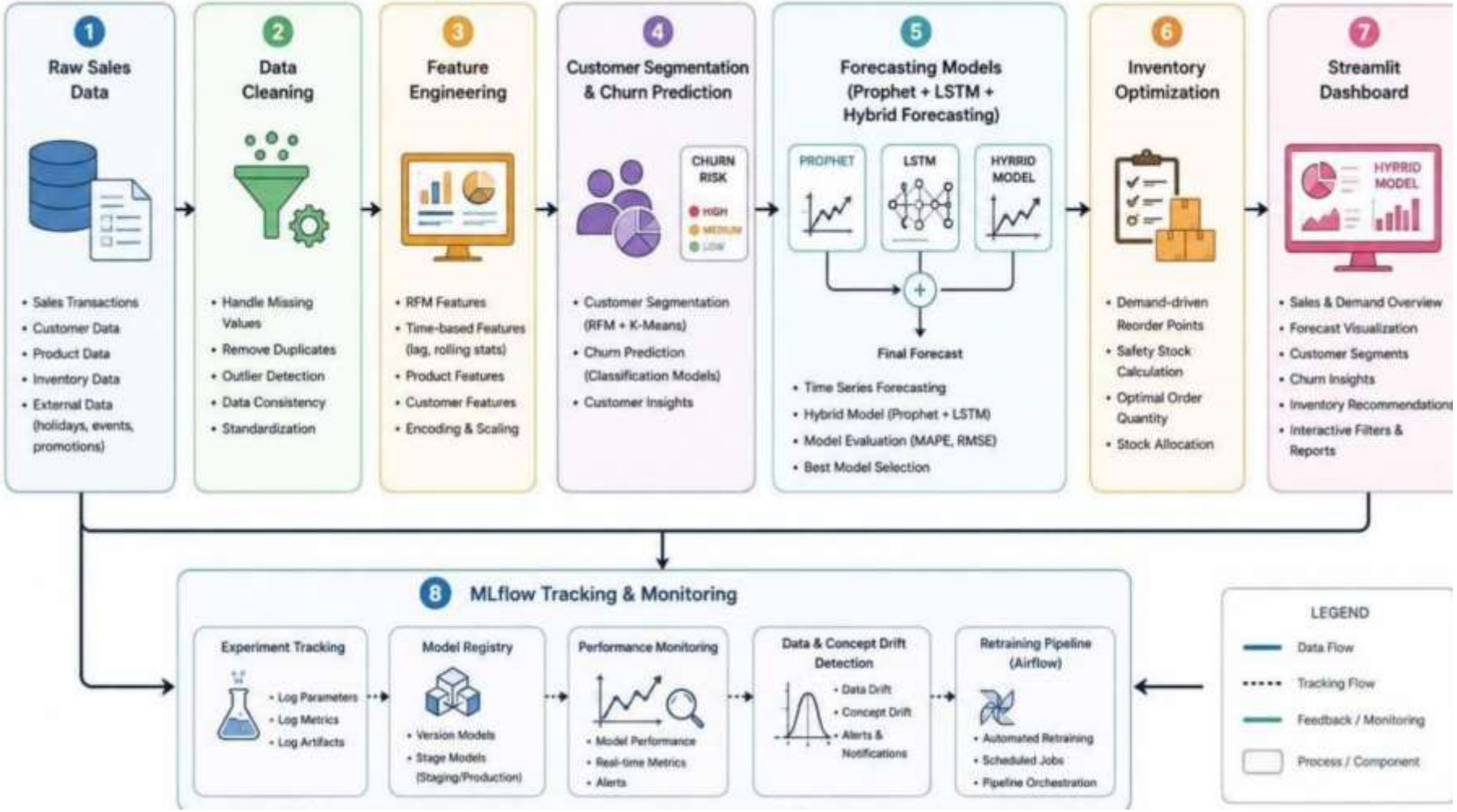
- Analyze sales performance and revenue trends.
- Segment customers with RFM analysis.
- Visualize demand trends and sales forecasts.
- Monitor inventory levels and stock issues.

Business Value

- Clearer view of sales and revenue performance.
- Better insight into customer purchasing behavior.
- Stronger demand planning and forecasting.
- Less manual reporting, with centralized analytics.

AN END-TO-END AI AND MLOPS WORKFLOW

Raw sales data becomes forecasts, inventory decisions, and continuously monitored models.



FOUR WEEKS FROM REQUIREMENTS TO DEPLOYMENT



Week 1

20 Sep – 27 2026
Planning, dataset collection and preprocessing, EDA



Week 2

28 Sep — 4 Oct 2026
Feature engineering, RFM segmentation, churn model



Week 3

5 Oct – 11 Oct 2026
Prophet and LSTM forecasting, hybrid model, inventory optimization



Week 4

12 Oct – 20 Oct 2026
Streamlit dashboard, Docker, deployment and testing, final report

HYBRID MODELING AND MLOPS SOLVED THE HARD PARTS

MODELING

Multiple views of the customer and the future

RFM and clustering segmentation, ML churn prediction, Prophet + LSTM hybrid forecasting.

MLOPS

Tracking, tuning, detection, orchestration

MLflow tracking, Optuna tuning, drift detection, Airflow orchestration.

CHALLENGES

The hard parts were operational

Inconsistent sales data, integrating multiple forecasting approaches, deployment configuration, Docker optimization.

SOLUTIONS

A modular path from data to decisions

Preprocessing pipeline, hybrid approach, modular architecture, automated workflows.

EIGHT FEATURES, EACH WITH A CLEAR ACCEPTANCE TEST

ID	Feature	Description	Acceptance Criteria
F1	Customer Segmentation	Groups customers based on behavior	Customers categorized successfully
F2	Churn Prediction	Predicts at-risk customers	Accurate churn insights generated
F3	Demand Forecasting	Predicts future demand	Reliable forecast trends shown
F4	Hybrid Forecasting	Combines multiple models	Improved prediction quality
F5	Inventory Optimization	Optimizes stock levels	Reduced overstocking risk
F6	Dashboard Analytics	Interactive visualization	Real-time insights accessible
F7	Drift Detection	Detects data changes	Alerts for model degradation
F8	MLflow Tracking	Tracks experiments	Performance history maintained

PYTHON STACK, DEPLOYED WITH STREAMLIT CLOUD AND DOCKER

Stack

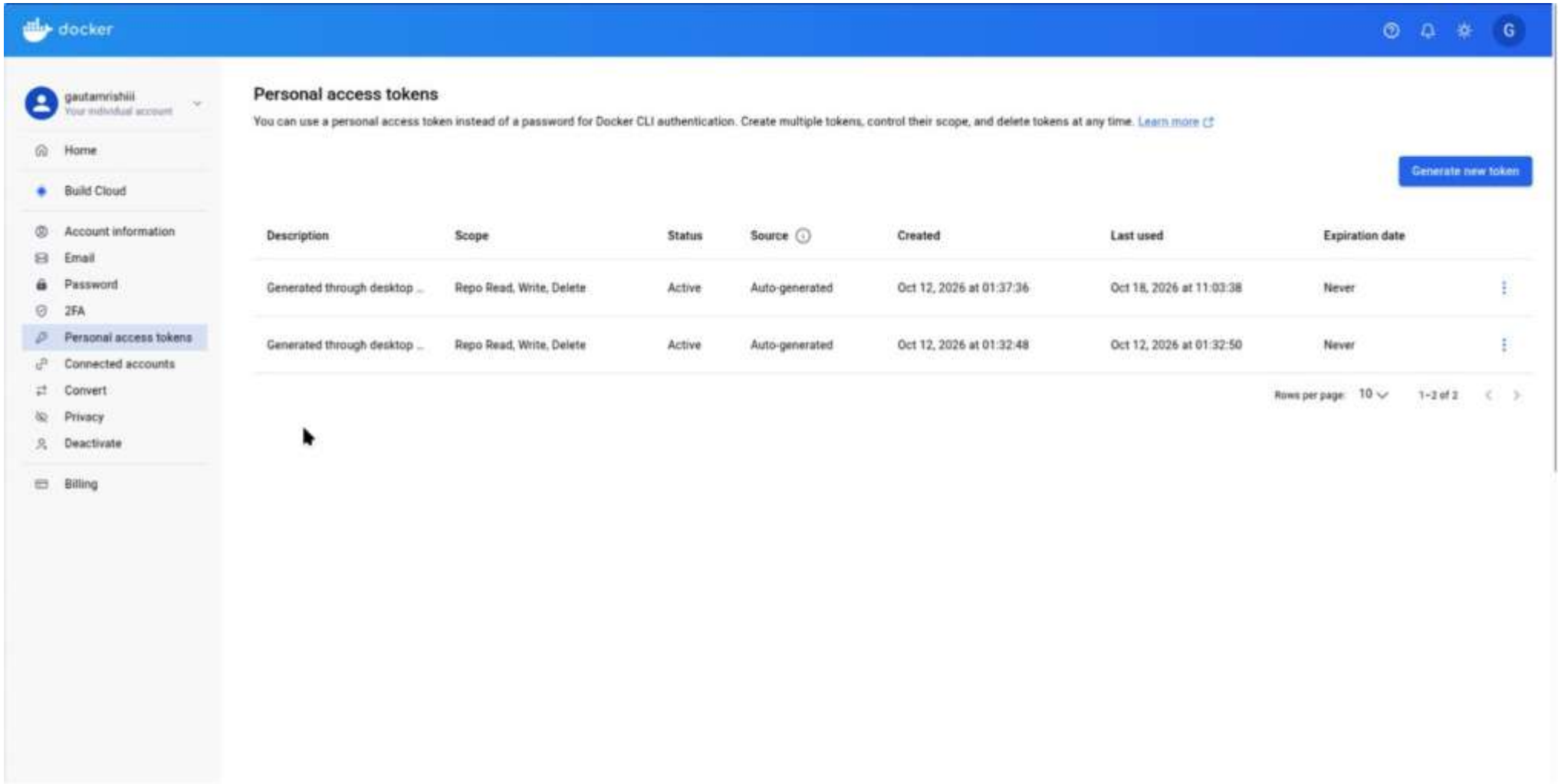
Python, Streamlit, Scikit-learn, TensorFlow, Prophet, LSTM, MLflow, Optuna, Airflow, Docker, Git & GitHub

Monitoring

Drift detection, model performance tracking, MLflow experiments

Deployment

Streamlit Cloud hosting, Docker containers, GitHub version control and CI/CD



THE DASHBOARD FORECASTS DEMAND AND RECOMMENDS STOCK

Home screen, forecast trend with weekly rhythm (Saturday peak), forecast vs adjusted demand (average 28,461) and inventory recommendations (average stock 389,789, minimum 319,939).



Dashboard Home Screen



Inventory Dashboard

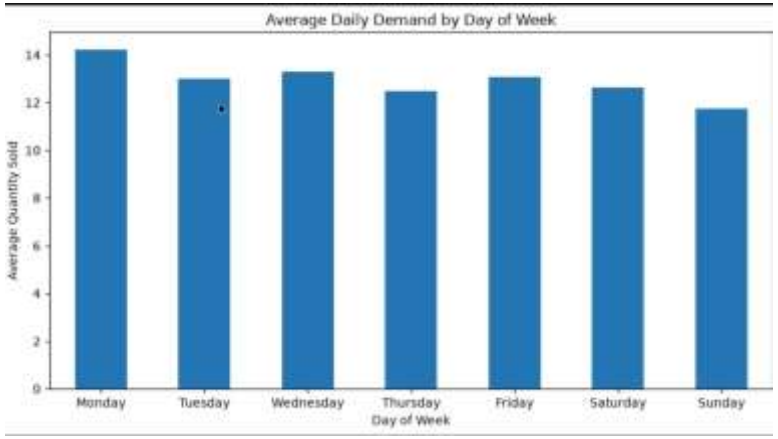


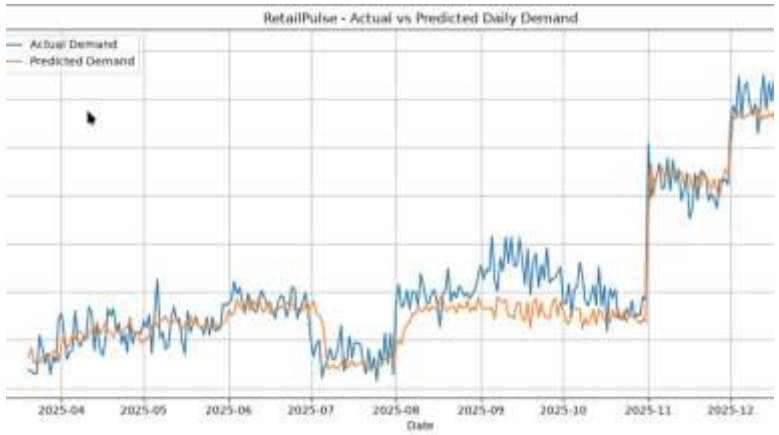
Forecast dashbord

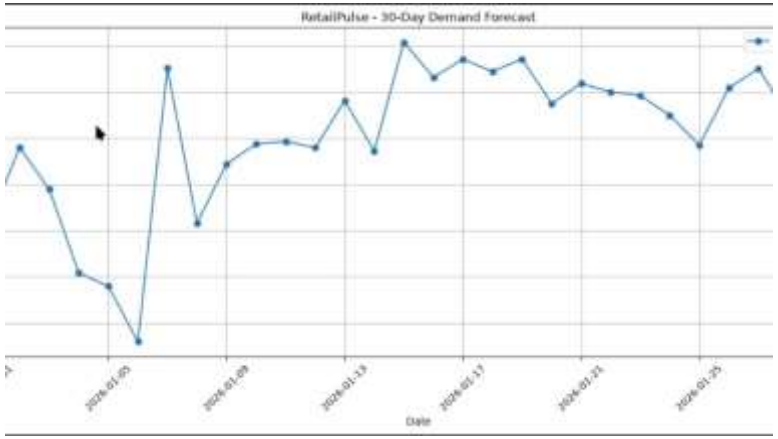


Customer Segmentation

RECENCY DRIVES CHURN, AND DRIFT ALERTS KEEP MODELS HONEST







Average Daily Demand by Day of Week

Average daily demand is highest on Monday at around 14.2 units, while Sunday records the lowest demand at approximately 11.7 units. Demand remains relatively consistent throughout the week, with a slight decline toward the weekend, which can help retailers plan inventory and stock replenishment.

Actual vs Predicted Daily Demand

The graph shows that predicted demand generally follows actual demand trends, capturing major fluctuations and seasonal changes throughout the year. However, the model underestimates some demand peaks, especially in November and December, indicating opportunities to improve forecasting accuracy.

30 Day Demand Forecast

The 30-day demand forecast predicts daily retail demand of approximately 11,080 to 11,400 units, showing moderate fluctuations throughout January 2026. These forecasts help retailers anticipate future demand, plan inventory levels, and make better stock replenishment decisions.

LEARNINGS, CONCLUSION AND WHAT COMES NEXT

Hands-on ML workflows, end-to-end deployment, Streamlit dashboards, forecasting and churn analysis.

Best practices: modular code, GitHub, MLflow tracking, Dockerized deployment.

CONCLUSION

RetailPulse demonstrates how data analytics can help retail businesses make better decisions by analyzing sales performance, customer behavior, demand patterns, and inventory levels. The project combines data analysis, RFM-based customer segmentation, demand forecasting visualizations, and inventory analytics in an interactive dashboard developed using Python, Streamlit, and Plotly.

The application is deployed using Docker and Render, making it accessible online. Overall, this project provided practical experience in data analysis, dashboard development, and deployment while demonstrating how data-driven insights can support better retail planning, customer understanding, and inventory management.

ROADMAP

Real-time analytics, cloud-native deployment, recommendation systems, enhanced automation and monitoring.